PRODUCTIVITY, RESTRUCTURING, AND THE GAINS FROM TAKEOVERS

by

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CES 13-18 April, 2013

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Abstract

This paper investigates how takeovers create value. Using plant-level data, I show that acquirers increase targets' productivity through more efficient use of capital and labor. Acquirers significantly reduce capital expenditures, wages, and employment in target plants, though output is unchanged. Acquirers improve targets' investment efficiency through better capital reallocation. Moreover, changes in productivity help explain the merging firms' announcement returns. The combined announcement returns are driven by improvements in target's productivity. Targets with greater productivity improvements receive higher premiums. These results provide some first empiricalevidence on the relation between productivity and stock returns in the context of takeovers.

Keyword: Takeovers, Announcement returns, Productivity, Investments, Wages, Employment

JEL Classification G34, D24

∗This paper is based on various chapters of my dissertation at the University of Michigan. I am indebted to members of my dissertation committee Amy Dittmar (co-chair), Jan Svejnar (co-chair), Charlie Brown, Francine Lafontaine, Amiyatosh Purnanandam, and Uday Rajan for their guidance and support. Suggestions provided by Ken Ahern, Angie Low, and an anonymous referee greatly improved this paper. I also thank Sugato Bhattarcharyya, Jagadeesh Sivadasan, Xianming Zhou, and seminar participants at the University of Michigan, University of Maryland, Cheung Kong Graduate School of Business, Peking University, Fudan University, and Shanghai Jiaotong University for helpful comments. The research was conducted when I was a special sworn status researcher of the US Bureau of the Census at the Michigan Research Data Center. Any opinions and conclusions expressed herein are mine and do not necessarily represent the views of the US Census Bureau. All results have been reviewed to ensurethat no confidential information is disclosed. This research is funded in part by Ewing Marion Kauffman foundation. Support for this research at the Michigan RDC from NSF (ITR-0427889) is also gratefully acknowledged. All errors are my own.

1 Introduction

Though it is well documented that takeovers increase the combined announcement returns for targets and acquirers (Andrade, Mitchell, and Stafford, 2001), focusing solely on stock market reaction to takeover announcements does not identify the sources of such gains. To shed light on the sources of gains from takeovers, several studies examine changes in plant-level productivity. For instance, Maksimovic and Phillips (2001) show that takeovers are followed by improved productivity.[1] These studies, however, do not attempt to identify the detailed mechanisms that lead to these changes in productivity. Yet identifying these mechanisms is important to understand which factors influence a merger's success or failure. In addition, little empirical evidence exists on the direct relation between announcement returns and productivity changes. Determining whether such a relation exists is critical because, if it does, it suggests that the stock market is not a sideshow, but rather it embeds information about underlying efficiency changes. The goal of this study, therefore, is to uncover the sources of productivity gains from takeovers and to relate them to the cross-sectional differences in announcement returns.

In this paper, I show that acquirers increase the productivity of targets through more efficient use of capital and labor and that the combined firms' announcement returns reflect these underlying efficiency improvements. I conduct the analysis on 1,430 mergers completed between 1981 and 2002, using data from the Securities Data Company (SDC) Mergers and Acquisitions database and plant level data from the US Census Bureau. These data allow me to identify detailed changes in plant-level output and input from before to after a takeover.

I analyze the channels through which productivity gains are created post-takeover. Because an increase in productivity, measured by total factor productivity (TFP), requires a relative increase in output compared with inputs, I consider changes to output vis-à-vis input following a takeover. The results show that, relative to comparable plants, capital expenditures, wages per worker, and employment all experience substantial declines while output remains constant. Essentially, the acquired plants are able to produce the same amount of output using less input. I find that employee layoffs are concentrated mostly among nonproduction workers in target plants, which supports the hypothesis that acquirers reduce management slack and transfer their own management know-how. Also, investment efficiency, measured as the sensitivity of targets' capital expenditures to investment opportunities, rises

[1] Other studies, including Lichtenberg (1992), McGuckin and Nguyen (2001), Yang (2008), Siegel and Simons (2010), Maksimovic, Phillips, and Prabhala (2011), and Maksimovic, Phillips, and Yang (2013) find similar results.

significantly after a takeover. These increased investment efficiencies are mainly achieved by acquirers that were conglomerate firms and that already had an active internal capital market prior to a takeover. These results indicate that increases in productivity stem primarily from acquirers' more efficient use of targets' capital and labor.

Building on these findings, I relate targets' productivity changes to the combined firms' announcement returns. The literature on corporate control proposes that wealth gains to shareholders reflect future efficiency improvements (Jensen and Ruback, 1983). Following this argument, the combined firm's announcement returns, which measure the total synergy gains, should capitalize at least in part the acquired firm's post-takeover productivity improvements. I construct a variable ΔTFP to measure the target firms' pre- to post-takeover percentage changes in TFP and regress the combined announcement returns on ΔTFP.

My analysis uncovers a statistically and economically significant effect of the target firm's ΔTFP on the combined announcement returns. Ceteris paribus, compared with acquiring a target at the 25th percentile of ΔTFP, acquiring a target at the 75th percentile increases the combined three-day announcement returns by 1 percentage point. If the target's pre-takeover market value is half of the acquirer's, such an interquartile change increases the combined returns by about 2.5 percentage points. The economic significance of the point estimates is considerable given that the sample median of the combined returns is about 3.1%. This evidence indicates that increases in the acquired firm's productivity are an important driver of the total synergy gains.

I next investigate the gains to targets and acquirers separately. Prior literature has consistently shown that targets receive high offer premiums from acquirers (Moeller, Schlingemann, and Stulz, 2004). Unfortunately, studies have yet to find specific economic gains that warrant such high premiums (Betton, Eckbo, and Thorburn, 2008). Many researchers attribute the high premiums to either behavioral distortions (Roll, 1986; Malmendier and Tate, 2008; and Baker, Pan, and Wurgler, 2012) or agency conflicts (Wulf, 2004; and Hartzell, Ofek, and Yermack, 2004). In this study, I hypothesize that the target firm's expected post-takeover improvements in productivity enhance its bargaining power in merger negotiation and that the expected post-takeover improvements in productivity materialize. The empirical evidence indicates that target firms with greater post-takeover productivity improvements receive higher premiums from acquirers. Other things being equal, an acquirer pays a target with ΔTFP at the 75th percentile about a 10 percentage points higher offer premium than a target at the 25th percentile. This result is consistent with insights from Rhodes-Kropf and Robinson (2008), who

emphasize the target's expected future surplus in determining bilateral bargaining positions. Lastly, I look at acquirers' own announcement returns. The results show that targets' improvements in productivity have a statistically and economically significant effect on acquirers' announcement returns only when the target is relatively large compared with the acquirer. Other things equal, if the target is half the size of the acquirer, an interquartile increase in target's ΔTFP is associated with 2 percentage points increase in acquirer's three-day announcement returns. This is a significant amount considering that the sample median of the acquirer returns is almost zero. Overall, these results indicate that announcement returns embed information about the acquired firm's post-takeover improvements in productivity.

This paper contributes to two related strands of literature on corporate takeovers. First, it builds upon studies on how takeovers affect targets' factor productivity. Maksimovic and Phillips (2001) find that acquired plants' productivity experiences greater improvements when the acquirer is more productive than the target. Schoar (2002) shows that acquired plants increase productivity when the acquisition is a diversifying transaction. More recently, Maksimovic, Phillips, and Prabhala (2011) present evidence that acquirers selectively choose target plants to retain and that only retained plants increase in productivity. Maksimovic, Phillips, and Yang (2013) compare productivity outcomes between on- and off-wave mergers and between acquisitions made by public and private firms. While these studies focus on the differences in productivity outcomes following different types of acquisitions, my approach is to peer inside the black box to identify how changes to labor and investments lead to increases in productivity. My findings concur with those in Devos, Kadapakkam, and Krishnamurthy (2009) who show that operating synergies come primarily from cutbacks in investment expenditures. According to my results, cutbacks in investments are just a first step, as I further show that targets' post-acquisition investments are more responsive to measures of investment opportunities than pre-acquisition. Moreover, conglomerate acquirers and acquirers that had an active internal capital market are better able to realize such investment efficiency gains. These results lend further support to the bright side view of conglomerate firms' internal capital market (Stein, 1997; and Maksimovic and Phillips, 2007, 2008).

Second, this study adds to the literature on the cross-sectional variation of merging firms' announcement returns. Most existing studies attempt to regress announcement returns on various proxies for economic gains such as Tobin's Q (Lang, Stulz, and Walkling, 1989; and Servaes, 1991), insider and analyst forecasts of synergies (Houston, James, and Ryngaert, 2001; and Devos, Kadapakkam, and

Krishnamurthy, 2009), operating cash flow (Healy, Palepu, and Ruback, 1992), corporate governance measures (Masulis, Wang, and Xie, 2007; and Wang and Xie, 2009), and product market synergy (Hoberg and Phillips, 2010). My findings support the conclusion that wealth gains to shareholders reflect expectations of improved operating performance. The crucial difference is that I regress announcement returns directly on a measure of plant-level productivity. Adopting this approach has two key advantages. First, plant productivity is a more fundamental determinant of economic efficiency. Second, plant level productivity measures are less subject to noise in accounting information surrounding mergers. This approach is similar to the one in Schoar (2002), who shows that stock market values track plant-level TFP in both conglomerate and single-segment firms. To the best of my knowledge, this is the first study that explores the relation between stock returns and plant-level productivity in the context of takeovers.

The rest of the paper is structured as follows. In the following section, I present the research objectives and develop some testable hypotheses from prior literature. Section 3 describes the data, sample, and variables. Section 4 analyzes how takeovers affect targets' inputs and outputs. In Section 5, I examine possible determinants of merger announcement returns from the perspective of productivity changes. Finally, Section 6 concludes.

2 Hypothesis development and related literature

In this study, I attempt to address two questions: (1) How do takeovers create productivity gains? (2) Do announcement returns for the combined firm reflect these productivity gains? In this section, I discuss the prior literature related to these two questions and develop some hypotheses to guide the empirical analysis.

2.1 *Sources of efficiency gains*

Previous studies have consistently shown that takeovers improve target firm's plant-level productivity (Lichtenberg, 1992; Maksimovic and Phillips, 2001, 2002; and Maksimovic, Phillips, and Prabhala, 2011). The goal of my study is to explain how takeovers achieve productivity improvements. An increase in productivity, measured by TFP, requires a relative increase in output compared with inputs. After acquisitions, acquirers have strong incentives to restructure the target firm to improve efficiency. Labor expenses and capital expenditures account for a large fraction of a firm's outlay of financial

resource. Meanwhile, corporate decisions on these two expenditures are prone to agency problems. Bertrand and Mullainathan (2003) show that chief executive officers (CEOs) prefer a "quiet life" by appeasing workers with excess wages. The Jensen (1986) free cash flow theory postulates that CEOs sacrifice firm's profitability for size by investing in unprofitable projects. Therefore, adjusting wages, employment, and investments not only reduces costs, but also restores productive efficiencies. I thus propose the following hypothesis to identify mechanisms through which productivity improvements occur.

Hypothesis 1. Acquirer reduces target's wage rate, employment, and investments, but the amount of output does not decline proportionally.

Lichtenberg and Siegel (1990) show that ownership changes are followed by substantial reductions in administrative overhead. Such actions are known to create values in leverage buyouts in the 1980s (Shleifer and Summers, 1988). Similarly, Kaplan (1989) finds that capital expenditures declined substantially in a sample of firms involved in management buyouts in the 1980s. Yet Servaes (1994) finds little evidence of investment cutbacks in a sample of around seven hundred firms that went private from 1972 to 1987. Based on analyst forecasts, Devos, Kadapakkam, and Krishnamurthy (2009) conclude that cutbacks in investment expenditures account for the majority of the operating synergies. All these studies focus on the levels of investments. In this study, I examine the efficiency of investments measured by the sensitivity of investments to investment opportunities.

Hypothesis 1 implies that acquiring firms create efficiency gains by correcting existing inefficiencies such as redundant employment, excessive wages, and inefficient investments. The necessary condition for these actions to be efficiency-enhancing is that the amount of output does not drop proportionally. An alternative to Hypothesis 1, as shown in McGuckin and Nguyen (2001), is that acquirers scale up the use of existing inputs and produce more outputs.

Hypothesis 2. Acquirer increases target's wage rate, employment, and investment, and the amount of output increases more than proportionally.

Essentially, Hypothesis 1 predicts that, after acquisitions, acquirers economize and downsize targets' inputs while maintaining or even increasing the outputs, and Hypothesis 2 predicts that acquirers maintain or even expand targets' scale of operations and produce proportionally greater outputs.

2.2 *Determinants of combined announcement returns*

Hypothesis 1 and Hypothesis 2 are concerned with how takeovers affect targets' productivity. I next examine how these productivity changes relate to the stock market valuation of the merger. Prior literature has consistently shown that mergers increase the combined market value for target and acquirer in the announcement period. To the extent that the stock market anticipates these activities, stock price reactions to merger announcement should capitalize these economic gains and losses. Therefore, the following hypothesis is proposed.

Hypothesis 3. Post-takeover improvements in target's productivity increase the combined announcement returns.

A large literature has examined the likes of Hypothesis 3 by regressing announcement returns on some proxies of economic gains. Using Tobin's Q as a proxy for managerial performance, Lang, Stulz, and Walkling (1989) and Servaes (1991) find that total takeover gain is higher when a well-managed (high Q) firm acquires a poorly managed (low Q) target. Masulis, Wang, and Xie (2007) and Wang and Xie (2009) present similar evidence using corporate governance measures. From a sample of the 50 largest mergers between 1979 and 1984, Healy, Palepu, and Ruback (1992) find a positive correlation between post-merger increases in operating cash flows and announcements returns. Based on a text-based measure for production market competition and differentiation, Hoberg and Phillips (2010) show that mergers create more value when the potential product market synergy (creating new products that increase product differentiation) is greater. Houston, James, and Ryngaert (2001) show that merger gains are closely related to acquiring firm's synergy forecasts in a sample of 41 banking mergers. While all these studies conclude that gains to shareholders reflect efficiency improvements, their approaches are subject to the criticisms that these variables (Q, governance, cash flow, product market competition, and synergy forecasts) represent at best noisy measures of efficiency gains.

2.3 *Determinants of offer premiums*

I next turn attention to the offer premiums that the targets receive. A robust empirical finding in the literature is that target firms' shareholders receive high premiums and, therefore, capture most of the joint gains.[2] For example, Moeller, Schlingemann, and Stulz (2004) report that targets receive premiums of about 65% and acquirers' three-day announcement returns are only about 1%. Little is known about what economic factors, if any, warrant such high offer premiums. Research to date

[2] Ahern (2012) provides contrary results based on division of merger gains in dollar value.

mostly attributes the high premiums to some behavioral or agency distortions. One influential view by Roll (1986) postulates that high offer premiums result from the acquirer CEO's hubris. Even if the acquiring firm can benefit from acquiring the target, the acquiring firm overpays for these benefits due to excessive self-confidence. Baker, Pan, and Wurgler (2012) find that acquirers set offer prices using the target's highest price during the 52 weeks before merger as a reference point. The target firm's agency conflicts can also affect offer premiums. Wulf (2004) and Hartzell, Ofek, and Yermack (2004) both suggest that target's CEO trades private benefits in exchange for lower premiums.

These behavioral- and agency-based explanations for offer premiums are in contrast to an efficiency driven explanation in which target firms bargain for more of the joint gains. Rhodes-Kropf and Robinson (2008) argue that the complementary nature of target's asset to acquirer's asset and target's expected future gains determine its bargaining position. If I assume that targets' realized productivity improvements are correctly anticipated by the target firm ex ante, targets with greater ex post productivity improvements are in a relatively stronger bargaining position and can, therefore, demand higher premiums. I propose the following hypothesis to examine whether offer premiums can be explained by target's post-takeover productivity gains.

Hypothesis 4. Targets with greater post-takeover productivity improvements receive higher offer premiums from acquirers.

3 Data sample and variable construction

I use plant-level data from the US Census Bureau to test the above hypotheses. The main data used in this paper come from a linked sample between the Annual Survey of Manufactures (ASM), the Census of Manufactures (CMF), and the SDC Mergers and Acquisitions database. This section describes the sample.

3.1 *Identifying mergers and acquisitions*

The data on mergers and acquisitions are taken from the SDC's US Mergers and Acquisitions database. Only deals completed between January 1, 1981 and December 31, 2002 and completed within 360 days of the announcement are included.[3] The transaction value of the deal is restricted to be at least $1

[3]I restrict the deals to be completed by 2002 because ASM and CMF data were available until 2005 when this paper was written and I want to follow the plants three years after the merger. I require the deals to be completed within 360 days so that I can identify the merger completion year cleanly.

million and the targets must be publicly listed with stock prices available from the Center for Research in Security Prices (CRSP). Acquirers must own less than 50% of the target's shares before and 100% after the merger. Acquirers can be public, private, or subsidiaries of public or private firms. I then match these targets to firms in the ASM and CMF database. A total of 1,430 matched target firms constitute the main sample used in this study.

Table 1 displays the composition of the sample across time and describes the basic characteristics of these deals. The time period 1981 to 2002 covers two merger waves. The number of transactions begins to increase in the 1980s, peaks in the late 1980s, and declines in the early 1990s, before picking up again toward the end of the sample period. The average deal value for the sample is large due to sample restrictions (publicly listed target manufacturing firms), about $950 million. The average three-day announcement period abnormal return for target firms is fairly stable, at about 25% throughout the sample period. A little more than half of these target firms (770) are acquired by another publicly-listed firm with information available from the CRSP. On average, the sample mean of the three-day announcement abnormal return for the acquiring firm is around zero. These figures buttress the well-established finding that mergers clearly benefit target shareholders while, on average, the acquiring firm's shareholder do not lose. The median value of the weighted sum of the announcement returns for target and acquirer combined is about 3.1%. This seems to suggest that these mergers create value to shareholders as a whole.

3.2 *Matching mergers to establishment data*

Plant-level information comes from ASM and CMF. These databases include information on total value of shipments, expenditures on intermediate and primary inputs, and other input measures such as materials, energy, and electricity costs. They also provide detailed information on wage and employment, which are covered only sparsely in Compustat. The ASM and CMF databases also circumvent the problems with accounting data, which are affected by managerial decisions especially after a merger. The economic census takes place in years ending with "2" and "7" and covers approximately 350,000 manufacturing plants each time. The ASM typically samples about 50 thousand plants in non-census years. Some 40 thousand other plants are selected with probability proportional to a composite measure of their sizes. Once a plant is surveyed, the ASM continues surveying this plant to form a five-year panel.

Once I identify a target firm, I use the firm-plant link to identify all the plants owned by the target

one year prior to a merger. Plants surveyed by the Census Bureau have a permanent establishment identifier (LBDNUM), which remains invariant to ownership changes. This allows me to follow the same plants before and after takeovers. I choose a three-year window, centered around the year when the merger is completed, to track the dynamics of these target plants.

3.3 *Variables*

In this subsection, I sketch a simple Cobb-Douglas production function to describe the variables used in subsequent estimations. Assume that a plant uses three factors (capital, labor, and materials) to produce output Y:

$$Y = AK^{\alpha}L^{\beta}M^{\gamma} \tag{1}$$

where Y denotes the dollar value of output, A denotes a constant term measuring the efficiency with which the plant uses its inputs in production (total factor productivity), L denotes labor, K denotes capital, and M denotes material cost.

Let W denote total payroll, which equals the product of the number of employees and average annual wage per worker, and I denote capital expenditures. A plant's operating cash flow from operations can be expressed as

$$\Pi = Y - W - I - M \tag{2}$$

This cash flow variable differs from a typical cash flow measure because the data do not cover firm-level costs, such as interest expenses, advertising, and research and development. Nevertheless, changes in operating cash flow can be expressed as a function of pre- and post-takeover changes in productivity Δ A, which incorporates changes to the total payroll Δ W, capital expenditures Δ I, and material costs Δ M:

$$\Delta\Pi = f(\Delta A) + \epsilon \tag{3}$$

Assuming that merger announcement abnormal returns capitalize the future cash flow gains or losses as in Eq. (3) by a proper discount rate r, I can therefore link changes in TFP to changes in market value of the firm. Specifically, I use Eq. (4) to test whether changes in productivity are related to changes in announcement returns, denoted as ΔV.

$$\text{Assuming that} \quad \Delta V \sim \frac{\Delta\Pi}{r} \quad \text{then,}$$
$$\Delta V = f(\Delta A) + \epsilon \tag{4}$$

I use TFP to measure a plant's overall efficiency. I estimate the following regression separately for each three-digit standard industrial classification (SIC) industry, controlling for plant fixed effects. The estimated residual term ϵ_{it} is referred to as the fixed effects total factor productivity (FE TFP) measure.

$$\text{Log}(\text{Output}_{it}) = \alpha + \beta_1 \text{Log}(\text{Payroll}_{it}) + \beta_2 \text{Log}(\text{Capital}_{it}) + \beta_3 \text{Log}(\text{Material}_{it}) + f_i + \epsilon_{it} \qquad (5)$$

where i indexes plants; t indexes year; Output is measured by the total value of shipments, deflated by industry-level price index; Payroll is total salaries and wages, and Capital is the value of the capital stock at the beginning of the year. Material is the sum of the cost of materials, intermediate goods, energy, and electricity. f_i is the plant fixed effect. TFP, the residual from the above estimation, compares the actual amount of output a plant produces with a predicted amount of output. The predicted output is what the plant is expected to produce given the amount of inputs it uses. Because coefficients on capital, labor, and material inputs vary by industry, this specification allows for different factor intensities in different industries. TFP can be interpreted as the efficiency measure of a plant relative to other plants in the same industry.

One drawback of the Cobb-Douglas production function is that the elasticities of output with respect to individual inputs are restricted to be constant and the elasticity of substitution between inputs is restricted to be equal to one. As an alternative, I consider the following second order Translog specification to estimate TFP:

$$\text{Log}(Y_{it}) = \alpha + \sum_j \beta_j \text{Log}(X_{it}^j) + \sum_{jj} \beta_{jj} \text{Log}^2(X_{it}^j) + \sum_{j \neq k} \sum_k \beta_{jk} \text{Log}(X_{it}^j) \text{Log}(X_{it}^k) + f_i + \epsilon_{it} \qquad (6)$$

where X_{it}^j is the quantity of input j used in production for plant i for year t. X includes capital stock, wage bill, and material costs. The estimated residual term ϵ_{it} in Eq. (6) is referred to as the Translog TFP measure. As expected, these two TFP measures are highly correlated. [4]

3.4 *Descriptive statistics*

Table 2 reports summary statistics for several key variables for target firms and plants across seven years around the merger. The top panel presents the firm-level characteristics. A substantial sample

[4] I also examine two other measures of performance: profit margin (used in Maksimovic, Phillips, and Prabhala, 2011) and return to capital (used in Bertrand and Mullainathan, 2003).

attrition can be seen from the second column of the table. For instance, for the year immediately prior to the merger, denoted as T−1, I am able to match 1,430 target firms. However, only 1,305 of these firms exist the year when the merger is completed (T) and 1,275 continue to exist one year later (T+1). This means that, for 155 mergers, I cannot find any plant left in ASM and CMF at T+1 that was originally owned by the target firm. The plant counts in Panel B reaffirm this attrition. For example, of all 11,129 plants in T−1, less than 70% remain in the sample three years after the merger. A cross check with Longitudinal Business Database (LBD), which covers the entire private business sector and includes robust longitudinal identifiers, reveals that 16% of the plants are closed and 21% of the plants are sold to other firms. These numbers are similar to what Maksimovic, Phillips, and Prabhala (2011) report: Acquirers close 19% and sell 27% of the acquired plants within three years of acquisition.

Mean and standard deviation of the number of employees and output are also reported. Two opposing trends can be observed in average values for employment and output levels. While the employment level gradually declines from T−3 to T+3, output experiences a steady increase during the same period, which suggests that labor productivity is rising. The sample means of both FE TFP and Translog TFP exhibit steady improvements immediately after the completion of takeovers at T and continue to grow through T+3.

3.5 *Control group*

Acquirers do not choose target firms at random, which leads to a selection bias problem. Changes between pre- and post-merger depend on a variety of economy and industry factors, such as technology or demand shocks, or on a continuation of firm-specific performance trend. Therefore, it is important to select a control group close to the counterfactuals (Harford, 2005). My approach here is to choose a group of comparable plants for each target plant based on industry, size, and productivity one year prior to the takeover year as benchmarks. Once I identify the control plants, I use the permanent plant identifier to follow them before and after the merger year the same way that I track the actual target plants. The three-step procedure is as follows:

1 I begin with all plants available in ASM and CMF and create three equal-size groups according to the total number of employees (large, medium, and small plants) within each three-digit SIC industry.

2 In the year immediately before the takeover (T−1), I sort the FE TFP within each SIC-size cell. From the pool of plants that are not acquired the following year, I retain two plants immediately above and two immediately below in TFP than a target plant. By construction, each target plant has at

most four comparable plants as controls.

3 After identifying all the control plants at T−1, I follow these control plants backward at T−3 and T−2 and forward at T, T+1, T+2, and T+3.

Several observations lead me to adopt the above dimension-by-dimension matching method when constructing my control group.[5] I choose to match on three plant-level characteristics (industry, size, and TFP) because later I focus on plant-level productivity and related outcomes. I first match on three-digit SIC industry. Mergers are clustered in industries undergoing exogenous shocks thus, the industry-year distribution of the target firms is not random. It is more useful to compare productivity levels within a given industry. I next match plants by size because a plant chooses an optimal size given its productivity and, hence, productivity distributions depend on size. I last select control plants that are closest to a target plant in terms of TFP within each SIC-size cell. This is similar to the spirit of Barber and Lyon (1996) on constructing control sample based on comparable pre-event performance. Matching the productivity level immediately before the merger allows me to compare productivity outcomes for targets relative to a group of similar-productive non-target counterparts. I run a simple regression $\text{TFP}_{\text{T-1}} - \text{TFP}_{\text{T-3}} = \alpha + \beta\text{Target} + \text{Industry FE} + \text{Year FE} + \epsilon$ and find that β is statistically indistinguishable from zero. This means that pre-takeover TFP growth was not systematically higher for target plants, thus dispelling the concern that preexisting trends in TFP are driving the results.

4 Real effects of takeovers

In this section, I analyze the channels through which productivity gains are created by examining the detailed changes to targets in outputs, inputs, labor and investments.

4.1 *Effects on outputs and inputs*

To compare real outcomes of targets relative to controls, I perform a difference-in-differences analysis for target and control plants in an unbalanced six-year panel after dropping the merger completion year T.

$$Y_{it} = \alpha + \beta_1 \text{After}_{it} + \beta_2(\text{After}_{it} * \text{Target}_i) + \text{Plant FE} + \text{Year FE} + \epsilon_{it} \tag{7}$$

In Eq. (7), the dependent variable Y includes efficiency measures (FE TFP, Translog TFP),

[5] An alternative is to use a propensity score matching. One concern is that model fit for target selection is poor (Jensen and Ruback, 1983). See Li and Prabhala (2007) for a detailed discussion of the propensity score matching methods when the first stage model fit is poor.

outputs, and inputs. After is a dummy variable that equals one for T+1, T+2, and T+3 and zero otherwise. Target is a dummy variable that equals one if the target plant's parent firm is acquired at T and zero for control plants. I only include plants that have at least one observation both before and after the event in the analysis. The above research design is similar to a program evaluation regression. The target plants belong to the treatment group, the comparable plants not to be acquired belong to the control group, and the treatment event is a takeover. Thus, I am evaluating the effect of takeover by comparing the before and after outcome changes of the treatment group with that of the control group. The coefficient on the interaction term After * Target can be interpreted as the difference in real outcomes for targets relative to controls from three years before to three years after a merger. Standard errors are clustered at the firm level.

I first confirm that target plants exhibit higher growth in TFP than similar plants in my sample. On average, three years after a merger, the acquired plants exhibit 2.4 percentage points higher FE TFP and 1.8 percentage points higher Translog TFP. These numbers are comparable to what the literature has found. For example, Schoar (2002) reports a 0.4% increase and Maksimovic and Phillips (2001) report a 2% increase in industry-adjusted TFP three years following a takeover.

I next explore how the acquiring firms improve target plants' TFP. I do so by breaking down TFP into output and input components and estimating the impact of takeovers on these components separately.

Table 3 reports the estimation results. All the dependent variables shown are in logarithms. The coefficients on the interaction term (After * Target) are negative and statistically significant for all outcome variables except for output. Post-takeover, target plants produce 0.2% less (statistically indistinguishable from zero) output than the controls, but they do so by using 0.9% less material, 12% less capital expenditure, 0.5% lower annual wages per worker, and about 2.1% fewer workers. The findings that targets' workers are negatively affected are against the hope that the acquiring firms could share some of the productivity gains with their acquired employees. Instead, target firm's employees are adversely affected by the takeovers. The high adjusted R^2 is due to the inclusion of a large number of plant fixed effects. If I use three-digit SIC industry fixed effects instead of plant fixed effects, the adjusted R^2 is more than halved. However, using plant fixed effects is preferable because one can control for all time-invariant plant-specific observable or unobservable factors.

I also look at wage and employment outcomes separately for two groups of workers: production workers and nonproduction workers. Nonproduction workers in this context are those who work within

a plant. I do not have information regarding nonproduction workers employed at the headquarters or other non-manufacturing establishments such as sales offices or distribution centers. The real effects on nonproduction workers, therefore, are probably under estimated. As seen in Table 4, while average wages for the two groups experience minimal declines, the reduction in jobs is severe, especially among nonproduction workers. On average, non-production workers are cut by about 6%, compared with 1.4% for production workers. This finding is consistent with the Jensen and Ruback (1983) view that takeovers create competition for the right to manage resources and achieve efficiency by replacing the target firms' management teams.

The above findings suggest that TFP improvements result from the acquiring firm's more efficient use of various inputs. Post-takeover, target plants are able to produce the same level of output relative to control plants, but do so with less capital and material, lower wages, and fewer workers. Furthermore, the efficiency improvements are achieved by more efficient investment decisions and reduced corporate overheads. These results offer support for Hypothesis 1, that is, acquirers improve targets' productive efficiency through restructuring and downsizing.

One thing worth noting is that estimates for Table 3 rely on information for continuing plants irrespective of further ownership changes post-takeovers. Maksimovic, Phillips, and Prabhala (2011) find that acquirers selectively choose plants to retain. I also look at the acquirer's decision to shut down plants in my sample. The untabulated results show that, post-takeover, a target plant is about 6.9 percentage points more likely to be closed than a similar plant and less productive plants are more likely to be closed. I also use the firm-establishment links in the census data to distinguish between plants that are retained by acquirers and plants that are sold to other firms. In my sample, about 21% of the plants are sold by acquirers within three years following a takeover. When I estimate Eq. (7) separately for the plants that are retained by the acquirers and for plants that are sold to other firms, I find that the retained plants experience a 2.9 (2.7) percentage points higher growth premium in FE TFP (Translog TFP) than comparable plants, while the sold plants do not see statistically significant changes in productivity. This finding confirms the conclusion in Maksimovic, Phillips, and Prabhala (2011) that only the retained plants enjoy productivity improvements. Also, I find that most of the changes to outputs and inputs are driven by the retained plants as well. In sum, the acquirers mainly restructure the retained plants and improve their productivity, whereas the sold plants experience little restructuring from pre- to post-takeover.

4.2 *Effects on investment efficiency*

Among all three types of inputs, capital expenditures undergo the largest reduction. One could wonder how a plant can sustain productivity growth with lowered investments. In this subsection, I consider the mechanisms through which investment cutbacks can increase productivity. A basic prediction of investment theory is that firms with better investment opportunities should invest more. Following this reasoning, I examine the sensitivity of investment to measures of investment opportunities, e.g., industry Tobin's Q, from pre- to post-takeovers. Empirically, I regress a plant's normalized capital expenditures on Q in the following specification.[6]

$$\text{Investment}_{it} = \alpha + \beta_1 \text{After}_{it} + \beta_2 \text{Q} + \beta_3 (\text{After}_{it} * \text{Q}) + \text{Plant FE} + \text{Year FE} + \epsilon_{it} \tag{8}$$

In Eq. (8), Investment is capital expenditures divided by sales and Q is the median Q of all public firms in the plant's three-digit SIC industry. If the investments of targets are more responsive to Q after the acquisition, then β_3 should be positive. I estimate Eq. (8) for target and control plants separately and compare β_3 for the two samples. I also use the average investment rates in the industry as a proxy for investment opportunities due to measurement error problems with Q. Industry average investments can work under a simple assumption: If other firms are making substantial investments, it is arguably because investment opportunities are good in this industry.

Because Fazzari, Hubbard, and Petersen (1988) find that cash flow, in addition to Q, affects capital expenditures, I augment Eq. (8) with a plant's profits (scaled by sales) to proxy for cash flow. The specification is:

$$\begin{aligned} \text{Investment}_{it} &= \alpha + \beta_1 \text{After}_{it} + \beta_2 \text{Q} + \beta_3 (\text{Q} * \text{After}_{it}) \\ &+ \beta_4 \text{Profit}_{it} + \beta_5 (\text{Profit}_{it} * \text{After}) + \text{Plant FE} + \text{Year FE} + \epsilon_{it} \end{aligned} \tag{9}$$

Table 4 reports the results of estimating Eqs. (8) and (9). The first three columns report results for target sample and the last three for control sample. In the first column, the estimate of the Q coefficient, β_2, is positive, which suggests that investments are responsive to industry Q before takeovers. The coefficient estimate of the interaction term, β_3, is positive and statistically significant.

[6] Previous studies on investment efficiency and corporate spin-offs (Gertner, Powers, and Scharfstein, 2002; Dittmar and Shivdasani, 2003) use similar regressions and find firms' investment sensitivities to Q rise after spin-offs.

This point estimate implies that the plant's post-takeover sensitivity of investment to Q rises by about two-thirds. In contrast, β_3 estimate in the fourth column is not statistically significant, which indicates that sensitivity of investments to Q does not rise post-takeover for control plants. An F-test for equality of β_3 for target and control sample rejects the null hypothesis at 1% confidence level. This evidence indicates that targets' investments are more responsive to investment opportunities after takeovers and control plants' investments are not.

To get a sense of the magnitude of the difference in the sensitivity of investment to Q before and after the merger, consider the following calculations. The sample average of the industry Q values is 1.51 with a standard deviation of 0.24. Using the estimates from the first column of Table 4, before a takeover, a one standard deviation increase in industry Q from its mean of 1.51 to 1.75 would increase normalized investment by 0.004. Evaluated at the mean investment rate of 0.112, the model predicts an increase to 0.116. After the takeover, the effects of an increase in Q on investments are larger. A one standard deviation increase in industry Q increases the investment by 0.01, which at the mean level amounts to an increase of investment rate to 0.122. In this case, the same change in industry Q results in a 8.9% increase in the investment rate after the takeover.

The results in Column 2 show that targets' investments also move closely in line with mean industry investments post-takeover. Column 3 reports results from estimate Eq. (9) including measures of cash flows. These results indicate that plants' investment is sensitive to its own operating profit, but such relations do not differ around takeovers. The coefficient on Profits * After is a significant negative when I do not include Industry Q * After. However, the coefficient on Profits*After becomes insignificant when I include Industry Q * After. This pattern implies that post-takeover investments are no longer responsive to cash flows after controlling for investment opportunities. As a result, the finding that targets' investments are more responsive to investment opportunities post-takeovers is robust to different measures of investment opportunities and to additional control variables.

By comparing the sensitivities of investments to Q from pre- to post-takeover, I shed light on how cutbacks capital expenditures potentially enhance productivity for target plants after takeovers. However, some caveats exist about this result. Erikson and Whited (2000) cast doubt on the validity of using Tobin's Q to measure investment opportunities due to measurement errors. If one can assume that the severity of measurement error does not change over time, the conclusion is valid, because I am comparing the changes of investment sensitivity to Q from pre- to post-merger. Nevertheless, the results are robust to using mean industry investment rates as another proxy for investment opportunities. In

a critique of the literature of investment efficiency after corporate spin-offs, Colak and Whited (2007) show firms that choose to spin-off and divest are larger, more diversified, and subject to more serious problems of asymmetric information. One can argue that these same critiques also apply to firms in mergers. However, using carefully chosen comparable plants in a difference-in-differences regression analysis alleviates such concerns. In sum, I find that targets' investment sensitivity to Q increases after takeovers relative to control plants.

4.3 *Exploring heterogeneity*

A question that naturally arises is whether increases in investment levels and investment efficiency can be attributed to some characteristics of the merger and characteristics of the acquirer. I investigate this question closely in this subsection.

4.3.1 *Related versus diversifying mergers*

I first examine how much of the change to investments can be attributed to diversifying mergers relative to related mergers. For investment regressions, I do so by breaking down the interaction term After * Target into two separate terms: After * Target * Related and After * Target * (1-Related). For investment efficiencies, I perform a similar exercise by breaking down the interaction term After * Q into two separate terms After*Q*Related and After* Q * (1-Related) and estimate Eq. (8). Related is a dummy variable that equals one if acquirer and target share the same primary two-digit SIC. This is a difference-in-differences-in-differences specification and allows me to compare the investment changes from related mergers (the coefficient on After * Target(or Q) * Related) with that from unrelated mergers (the coefficient on After * Target(or Q) * (1-Related)).

I report the investment regression results in Table 5 and investment efficiency results in Table 6. As can be observed in Column 1 of Table 5, coefficient on After * Target * Related is -0.136, which is greater (in absolute value) than the coefficient on After * Target * (1-Related), -0.121. The comparison suggests that related mergers realize greater investment savings than unrelated mergers. This result is not surprising considering that merging similar lines of business can avoid duplicate investments.

I then turn attention to investment efficiency results in Column 1 of Table 6. The coefficient on After * Q * (1-Related) of 0.020 is significantly greater than the coefficient on After * Q * Related, 0.014. This comparison indicates that the overall improvements in investment efficiency are driven more by unrelated mergers than by related mergers, although targets acquired by related acquirers

experience greater investment cutbacks.

4.3.2 *Conglomerate versus single-segment acquirers*

One of the most examined questions among economists is whether firm boundaries matter for allocation of resources. I shed light on this issue by comparing the resource allocation in targets acquired by conglomerates versus single-segment firms. In my sample, about 64% of the mergers are undertaken by acquirers that were conglomerate firms before the acquisition.[7] How do they differ from single-segment acquirers?

In Column 2 of Table 5, I include interaction terms After * Target * Conglomerate and After * Target * (1-Conglomerate) to investigate whether the acquirer was a conglomerate pre-acquisition matters for target's post-acquisition investment rates. The results suggest that conglomerate acquirers are more aggressive in cutting targets' capital expenditures. Conglomerate acquirers reduce about 5 percentage points more targets' capital expenditures than their single-segment counterparts.

In Column 2 of Table 6, I add After * Q * Conglomerate and After * Q * (1-Conglomerate) to investigate whether acquisitions undertaken by conglomerates differ from single-segment firms in terms of enhancing investment efficiency. The results indicate that increases in investment efficiencies are confined to targets that are acquired by acquirers that were conglomerate firms pre-merger. The relative improvement in investment efficiency is economically large. Estimates in Column 2 suggest that a one standard deviation increase in Q results in 12.7% greater increase in the investment in targets acquired by conglomerate acquirers than single-segment acquirers. The above results show that acquirers' organization structure matters for resource reallocation, but through what mechanisms? I shed light on this question by examining the internal capital markets of acquirers.

4.3.3 *Do internal capital markets of acquirers matter?*

A large body of work has examined the merits of internal capital markets in conglomerate firms (see Maksimovic and Phillips (2007) for a summary). One view holds that conglomerates, through centralized control over the capital allocation process, can better allocate capital than single-segment firms (Stein, 1997). In contrast, the dark side view of internal capital market posits that the presence of corporate socialism in conglomerates impedes capital from going to its best use (Rajan, Servaes,

[7]The conglomerate firm here is defined as a firm with at least two segments (three-digit SIC) each of which accounts for more than 10% of a firm's total output according to Compustat Segment files or employs more than 10% of a firm's total employment according to LBD.

and Zingales, 2000).

I proceed to analyze whether the changes to investments acquired by conglomerate firms can be explained in part by the internal capital market (ICM) intensity. While the results above examine whether investment differences exist across firms acquired in different types of mergers, I now examine whether differences exist among conglomerate acquirers.

More concretely, in Columns 3 to 5 of Table 5 and Table 6, I focus attention only on cases in which the acquirer was a conglomerate before the merger and report estimates of a triple interaction term of After * Target(or Q) * ICM intensity. I employ three measures of ICM intensity suggested in the literature.

The first measure, Diversity, closely follows Rajan, Servaes, and Zingales (2000). It is defined as the standard deviation of the segment-asset weighted (imputed) Q, divided by the equally weighted average (imputed) segment Q. In particular, Diversity= $\frac{\sqrt{\sum_{j=1}^{n} \frac{(w_j Q_j - \overline{w_j Q_j})^2}{n-1}}}{\overline{Q}}$, where w_j is segment j's share of total sales and $\overline{w_j Q_j}$ is the average asset weighted Q_j, n is the number of segments and $\overline{Q}$ is the equally weighted average segment Q. As Rajan, Servaes, and Zingales (2000) point out, this measure seeks to encompass three distinct features of ICM: the number of different divisions, the correlation of investment opportunities between the divisions, and the size differences between these divisions. A higher value of the variable corresponds to scenarios in which the role of ICM is expected to be more important. In my sample, Diversity ranges from 0.09 to 1.93 with a mean value of 0.98.

The second measure, Reallocate, is obtained from Billett and Mauer (2003). This variable intends to capture the wedge between the cash surplus in some segments and the cash deficit in others. The degree of disparity in cash needs across segments is measured by the sum of absolute values of the difference between cash flows and capital expenditures across all segments. This is standardized by the absolute value of total investments less the total cash to account for the availability of total capital. Finally, the variable is normalized by the total sales of the conglomerate. Specifically, Reallocate= $\frac{\sum_{j=1}^{n} |I_j - CF_j| - |\sum_{j-1}^{n} (I_j - CF_j)|}{Sales}$, where Sales are the total sales of the conglomerate, I_j is segment j's investments, and CF_j is its cash flows. Hypothetically, ICMs become more important when there is a greater degree of mismatch between the cash inflows and outflows of segments. Therefore, a higher value proxies for greater importance of ICMs for the conglomerate. In my sample, Reallocate ranges from 0 to 0.53 with a mean value of 0.091.

Finally, the third measure of ICM importance is the inverse of Herfindahl index of sales of all

segments of the firm. Specifically, $\frac{1}{Herfindahl} = \frac{1}{\sum_{j=1}^{n}(\frac{Sales_j}{\sum_{j=1}^{n} Sales_j})^2}$, where $Sales_j$ is segment j's sales. A higher value implies that the potential competition for resources by different segments within a conglomerate is stiffer, where ICMs expect to play a more important role. In my sample, the inverse of Herfindahl index ranges from 1.08 to 9.12 with a mean value of 3.43.

In the empirical analysis, the triple interaction term of After * Target(or Q) * ICM intensity (Diversity, Reallocate, and inverse of Herfindahl) is intended to examine whether ICM intensity in conglomerate acquirers affects targets' investments and investment efficiencies after the acquisition. Regression estimates in Columns 3 to 5 in Table 5 suggest that acquirers that ran a more active internal capital markets pre-merger cut back more on targets' capital expenditures.

Similarly, in Columns 3 to 5 in Table 6, for all three proxies of ICM intensity, the coefficient estimates indicate that targets' post-merger investment sensitivities to Q are positively related to the pre-merger ICM intensity of acquirers. The economic significance is also substantial. For instance, the coefficients in Column 3 suggest that a change from 25th percentile to 75th percentile of the diversity of the acquirer leads to around 27% increase of investment sensitivities to Q for targets after the merger. In sum, investments become more efficient in targets acquired by conglomerate firms with active internal capital markets.

Conglomerate firms differ from single-segment firms in many other ways and acquisitions undertaken by conglomerate and single-segment firms are also different. Nonetheless, by presenting the effect of internal capital markets on investments, the above analysis shows that conglomerates with more active ICMs are more efficient in reallocating targets' capital post-acquisition. My results provide further support to the arguments in Maksimovic and Phillips (2002) that conglomerate firms primarily behave as value maximizers and their internal capital markets facilitate the efficient transfer of resources.

4.3.4 *Other characteristics of acquirers: size and market-to-book*

I also explore whether improvements in targets' investment efficiency vary with other characteristics of the acquirer. Being part of a larger organization could improve target's investment through better access to capital. Results in Column 6 in Tables 5 and 6 suggest that investments and investment efficiencies improvements are not directly related to acquirer's sheer size. Column 7 of Tables 5 and 6 introduces yet another acquirer characteristic, the market-to-book ratio, which is found by many studies to determine announcement returns. The coefficient on the triple interaction term is insignificant, which implies that target's investment changes cannot be explained by acquirers' market-to-book ratios. In

results not shown here, I find that targets' increases in investment efficiencies are weakly related to acquirers' corporate governance (as measured by the G-index) but do not depend on acquirers' financing constraints. All these variables do not materially affect the sign or significance of the ICM variables.

4.4 *Robustness checks*

I check the robustness of the results by performing two additional tests. First, I use a different control sample to mitigate the concern that the results are an artifact of a particular control sample. Second, I utilize a small sample of incomplete mergers that are ultimately withdrawn by acquirers to conduct a placebo test.

4.4.1 *A different control sample*

I construct a different sample of control plants by matching on industry, age cohorts, and total number of employees pre-takeover following Davis, Haltiwanger, Jarmin, Lerner, and Miranda (2008). Specifically, I create three equal-size age groups (young, middle-age, and old) within each three-digit SIC industry. In the year prior to the takeover, I sort the total number of employees with each SIC-age cell and retain two plants immediately larger and two immediately smaller than the targeted plant to form my control groups.

I perform the same differences-in-differences estimations using this new group of control plants. In results not reported here, I find that FE TFP (Translog TFP) growth premium for target plants over this new sample of control plants is about 1.7 (1.9) percentage points. In contrary, all inputs decline relative to controls. Capital expenditures decline by about 11% more than the controls. Wage rate drops by nearly 1% and employment level by about 2% more than controls. This evidence confirms that the previous estimates are robust to the control sample selections.

4.4.2 *A placebo test with unsuccessful mergers*

Several recent studies (see, e.g., Seru, 2010) use mergers that fail to consummate to identify the treatment effect of takeovers. I follow this approach and identify 236 potential target firms that bidders ultimately withdraw. The sample of unsuccessful mergers allows me to perform two tests. First, I use these unsuccessful targets as counterfactuals for how the successful targets would have affected productivity after mergers. In particular, I perform a difference-in-differences regression using these unsuccessful target firms as a new control sample. The untabulated results show that, relative to these

unsuccessful targets, target plants from completed mergers exhibit a 1.5 (1.8) percentage points gain in FE TFP (Translog TFP) post-takeover. In addition, I conduct a placebo test to examine whether these unsuccessful takeovers improve their potential target plants' TFP. My results (unreported) show that unsuccessful takeovers fail to increase their potential target plants' productivity relative to comparable plants. This finding further supports that takeovers improve productivity through the actual change of ownership instead of potential threats by announcements.

In sum, these results consistently show that target plants experience about 1.5 to 2 percentage points net productivity gains over control plants. Capital expenditures experience significant decline by about 10% to 12%, compared with controls. Employees fare worse after takeovers. Average wage and employment are lowered by about 1% to 2% relative to controls. The employment cuts are mostly concentrated among nonproduction workers and production workers are slightly affected.

4.4.3 *Effects in a longer span post-takeover*

One might worry that a three-year window might not fully capture the effects acquisitions exert on target plants. I extend the post-takeover window to five years, and I find the acquired plants continue to display higher growth premiums in productivity after the third year post-takeover. Relative to similar plants, the acquired plants, in their fourth and fifth year after a takeover, enjoy a 1.5 (1.8) percentage points higher growth rate in FE TFP (Translog TFP).

Factors of production, such as capital and labor, take time to adjust to shocks and are, therefore, dependent on the levels in the past. To account for the dynamic nature of capital and labor adjustment, I estimate a dynamic equation to analyze the effects of takeovers on inputs. The level of output for target plants remains on par with similar plants not only for the first three years but also for the fourth and fifth year after a takeover. In contrast, capital expenditures continue to decline in the fourth and fifth year post-takeover by more than 10%. Employment levels experience steady declines of about 1.6% throughout the five-year period after a takeover. Production workers and especially nonproduction workers experience substantial job losses for target plants relative to comparable plants during the five-year window after a takeover.

4.5 *Acquirers' incumbent plants*

While the focus of this paper is on target plants after a takeover, a related question is how the acquisitions affect acquirers' incumbent plants. Only a small portion of the acquiring firms can be

matched to ASM and CMF for this purpose. Of the 1,430 target firms, 770 of them are acquired by another US publicly listed firm. Among these acquiring firms, 641 of them can be matched to ASM and CMF. Many of these acquiring firms are repeated acquirers within a few years, which makes it difficult to assign "before" and "after" according to a takeover event. Based on a restricted sample of acquisitions that I can obtain a clean three-year post-takeover window, I examine how a major takeover affects acquirers' incumbent plants. I use the same method as in Subsection 3.5 to choose a similar group of control plants and estimate a difference-in-differences model. I find that, relative to similar plants, acquirers' incumbent plants also experience a net improvements of 2.4 (2.8) percentage points in FE TFP (Translog TFP). This result is at odds with the finding in Schoar (2002), who shows that productivity of acquirers' incumbent plants experience marginal (but statistically insignificant) declines. However, my findings are more in line with Maksimovic, Phillips, and Prabhala (2011), that acquirer's retained plants also increase productivity. This is not surprising given that my sample frame and estimation methods are similar to theirs.[8] I do not report these results here to save space.

5 Determinants of announcement returns

The above results show whether and especially how target plants achieve improvements in TFP after takeovers. In this section, I analyze the relation between what happens inside the target firm and what happens to stock returns when the merger is announced. In particular, I explore the connection between cross-sectional differences in announcement returns and the changes in target firm's productivity. To pursue this, I augment the standard announcement returns regressions with the target firm's change in FE TFP between T−1 and T+1. The specification is:

$$\text{Combined CAR} = \alpha + \beta\Delta\text{TFP} + \text{Deal Characteristics} + \text{Acquirer/Target Characteristics} + \epsilon \quad (10)$$

In Eq. (10), the dependent variable is the three-day cumulative abnormal returns (CAR) for target and acquirer combined. CAR is calculated using a market model estimated over the $(-205, -6)$ interval. Combined CAR is calculated as the market value weighted sum of target CAR and acquirer CAR. The combined announcement returns measure the total value to shareholders from the merger. Later, I consider target's offer premiums and acquirer CAR separately using similar specifications. I

[8] I obtain similar results as in Schoar(2002) when I simply look at before and after changes. I find acquirers' incumbent plants improve TFP relative to similar plants using a difference-in-differences estimation.

calculate the offer premium as the ratio of deal value to the target's market value 30 days before the merger announcement. The variable of interest is β. If β is positive, it implies that the announcement returns reflect the underlying productivity improvements.

I calculate ΔTFP by subtracting the target firm's TFP at T$-$1 from its TFP at T+1, in which a firm's TFP is formed as the output-weighted average of all the retained plants' FE TFP. This method of construction is designed to capture two aspects of changes: the improvement effects over the retained plants and the selection effects of divested and closed plants. Only target firms with at least one plant retained by acquirer at T+1 are included in the sample. Table 7 reports the mean, standard deviation, and pairwise correlation for some variables of interest. Among the 1,430 merger transactions, 1,226 target firms have non-missing values on ΔTFP and 660 target firms are acquired by another publicly listed firms with information available from CRSP. According to the pairwise correlation, ΔTFP is positively correlated with combined CAR, offer premiums, and as the acquirer CAR.

To explore other determinants of announcement returns, I mainly consider two groups of factors: Acquirer/Target firm characteristics and deal characteristics. The firm characteristics include firm size (measured by book value of assets in logs) and the market-to-book ratio. Both variables are measured at the fiscal year prior to the takeover announcement. Deal characteristics include method of payment (cash versus stock), target-acquirer industry relatedness, and whether a merger is a tender offer.[9] I create two variables *All Cash* and *Some Stock* to distinguish between mergers financed by cash and by stock. I measure target-acquirer industry relatedness using a dummy variable *Related* that equals one if target and acquirer share the same primary two-digit SIC industry. I create an indicator variable *Tender Offer* that equals one if the transaction is a tender offer and zero otherwise. I also create a relative size measure for the target, calculated as the share of the target firm's market value to the sum of the target and the acquirer's market value, as in Hoberg and Phillips (2010).

5.1 *Effects of productivity changes on combined CAR*

Table 8 reports estimates from ordinary least squares (OLS) regression of the combined CAR. The first column reports regression results when I only include ΔTFP and deal characteristics. I further add acquirer's and target's characteristics in columns 2 and 3, respectively. The results on ΔTFP are strengthened when I include all set of controls. Across all three columns, the coefficients on ΔTFP are positive and statistically significant, which suggests that improvements in productivity are

[9] Schwert (2000) discusses deal attitude, but SDC records only less than 5% of deals as "hostile" and the Bureau of Census restricts small sub-sample information disclosure.

associated with higher combined CAR. In terms of economic significance, the coefficient of 0.036 in Column 3 implies that acquiring a target with ΔTFP at the 25th percentile yields 0.9 percentage points lower combined CAR than acquiring a target with ΔTFP at the 75th percentile. This amount is considerable given that the sample median of the combined CAR is about 3.1%. These results provide support for Hypothesis 3 that improvements in the acquired firm's productivity are associated with higher announcement returns for the combined firm.

Results on other control variables are generally consistent with findings in previous literature. Related mergers generate about 1.7 percentage points higher combined returns than unrelated mergers. If the acquirer uses some stocks to pay for the target, the combined returns are lowered by about 2.7 percentage points. I find that large acquiring firms tend to make value-destroying acquisitions, similar to the results shown in Moeller, Schlingemann, and Stulz (2004, 2005). Prior studies find that the Tobin's Q (a close surrogate of market-to-book ratio) has an ambiguous effect on merging firms' announcement returns. For instance, Lang, Stulz, and Walkling (1989) and Servaes (1991) find that combined CAR is positively related to the acquirer's Q, while Moeller, Schlingemann, and Stulz (2004) show the contrary. In my sample, I find that the acquirers' market-to-book ratios are inversely associated (though statistically insignificant from zero) and targets' market-to-book ratios are positively associated with announcement returns for the combined firm.

In Columns 4 to 6, I further add an interaction term between RelaSize and ΔTFP. The idea is that some of targets' improvements in TFP might not affect the combined returns in a significant way because targets are generally small relative to acquirers. The variable RelaSize is statistically significant at 1% level. In fact, the adjusted R^2 is about 0.20 in combined CAR regression when I include this variable alone. It rises to 0.24 when I include RelaSize and ΔTFP, and it further rises to 0.28 when I include both variables and the interaction term. The results show that the interaction term RelaSize * ΔTFP is positive and statistically significant at the 1% level. When I include the interaction term, the variable ΔTFP itself turns out to be negative but insignificant. The economic significance of the coefficient on RelaSize and ΔTFP is substantial. Ceteris paribus, if a target's pre-takeover market value is half of the acquirer's, an acquisition of a target firm with ΔTFP at the 25th percentile yields about 2.5 percentage points lower combined announcement returns than a target with ΔTFP at the 75th percentile. Therefore, the positive effect of ΔTFP on combined CAR is more pronounced when target is relatively larger compared with acquirer.

5.2 *Effects of productivity changes on premium*

I next estimate a similar regression of offer premium on ΔTFP to test Hypothesis 4, that is, whether offer premiums can be explained in part by post-takeover productivity improvements. The results are presented in Table 9. In the first column, I include only deal characteristics and target's characteristics in addition to ΔTFP. The coefficient on ΔTFP is positive and statistically significant at 5% level, which suggests that the acquiring firm does pay a higher premium to a target with greater post-takeover improvements in productivity. The point estimate on ΔTFP is 0.319. To interpret the economic magnitude, other things being equal, a target at the 25th percentile of ΔTFP receives about 8 percentage points lower premium than a target at the 75th percentile. Such a change is sizeable considering that the median value of the offer premium in my sample is about 50%. Hence, Hypothesis 4 is supported. In the cross-section, target firms with greater post-takeover improvements in productivity receive higher offer premiums from the acquirer.

Other results are generally as expected. A target that shares the same primary industry with the acquirer receives a 30 percentage points higher premium than otherwise. The target is willing to trade for a significantly lower premium for cash payment. Tender offers are generally associated with higher premiums. Targets with higher market-to-book ratios receive lower premiums. Baker, Pan, and Wurgler (2012) advance a reference point explanation arguing that acquirer's offer prices are benchmarked against target's previous 52-week highest price. I confirm their findings and show that target's *52-W High Price* has a positive and statistically significant effect on offer premiums. This result is robust even after I control for target's highest price in the past 26 weeks and the past four weeks in some specifications.

Columns 2 to 6 in Table 9 report results when I control for *RelaSize*. The number of observation is smaller than in the first column because some acquirers are private firms without information on market values. In untabulated results, I find that a private acquirer pays a significantly lower premium than a public acquirer. The variable ΔTFP is still positive and statistically significant at the 5% level. The point estimates are similar to each other across different specifications. This indicates that targets with greater productivity improvements receive higher premiums not only from public acquirers but also from private acquirers. The negative sign on *RelaSize* implies that as the target increases in size relative to the acquirer, offer premiums drop. It is not clear why target's offer premiums are a function of relative size. Jarrell and Poulsen (1989) find similar evidence and conjecture that acquirers could offer lower premiums to a larger, more diffusely held firms. In Columns 4 to 6, I further add an

interaction term between target's relative size and ΔTFP. In contrast to the results shown in Table 8, this interaction term is not statistically significant at a conventional level, which suggests that the effect of ΔTFP on offer premium is not contingent upon target's relative size. In untabulated results, I find that the regression results using target's announcement returns are generally similar to the results on offer premiums. Schwert (1996) discusses the close relation between offer premiums and target announcement returns. I report results on offer premiums because offer premiums better capture the wealth gains to target firm's shareholders.

5.3 *Effects of productivity changes on acquirer CAR*

I last look at the acquiring firm's announcement returns. The results are reported in Table 10. ΔTFP has a positive effect on the acquirer's CAR, but the point estimate is not statistically significant at a conventional level. In fact, few regressors have significant explanatory power for the acquirer's CAR and the adjusted R^2 is very low despite the fact that I control for numerous variables. The two exceptions are variables for stock mergers and acquirer's size. Both variables are negatively associated with acquirer's CAR, which are consistent with the stylized facts presented in Moeller, Schlingemann, and Stulz (2004, 2005). Mergers undertaken by large acquirers and mergers financed using stocks are generally associated with negative announcement returns for acquiring firms.

Except for acquirer size and method of payments, prior literature has yet to find consistently any variables that can explain variations in acquirer's announcement returns. Houston, James, and Ryngaert (2001) argue that improvements over the target firms might not lead to discernable gains to the acquiring firms themselves because acquirers are usually much larger than targets. In Columns 4 to 6, I control for the interaction terms between target's relative size and ΔTFP. Results show that RelaSize * ΔTFP is positively associated with acquirer CAR. The point estimates on RelaSize * ΔTFP range from 0.097 to 0.116 and are statistically significant at the 5% level. It implies that improvements in target's productivity exert a positive and significant effect on acquirer CAR only when target is relatively larger compared with acquirer. Other things equal, if the target is half the size of the acquirer, acquiring a target at the 25th percentile of ΔTFP generates about 1 percentage points lower announcement returns than acquiring a target at the 75th percentile. The magnitude of such an increase is economically significant as the sample median of the acquirer CAR is around zero.

I already show that RelaSize * ΔTFP has a positive effect on combined CAR but no effect on offer premiums. This finding reveals that the effect of ΔTFP on combined CAR is mainly manifested through

ΔTFP's effect on acquirer CAR when the target is relatively larger compared with the acquirer. This finding further supports the hypothesis that the announcement returns are driven by post-takeover improvements in target's productivity.

5.4 *Breaking down the TFP effects*

The analysis in Section 4 reveals that acquirers improve targets' TFP by cutting back on wages, employment, and capital expenditures and by increasing investment efficiency. I now break down the overall effects of ΔTFP into these four components and examine the relation between announcement returns and each component. For each variable X = Wage, Employment, and Capital Expenditures, I calculate ΔX by subtracting the target firm's X at T−1 from its X at T+1, where a firm's X is formed as the output weighted average of all the retained plants' X. The last component is investment efficiency.

A firm invests more efficiently if it invests relatively more in its high-Q segments than its low-Q segments. I construct a variable of RINV (relative investment ratio), similar to the one used in Ahn and Denis (2004), to measure a firms overall investment efficiency. RINV is calculated as the sales-weighted sum of investments in high-Q plants minus investments in low-Q plants. Greater values of RINV indicate that the firm is investing relatively more in plants in high-Q industries than plants in low-Q industries. In particular, RINV is constructed as,

$$RINV = \frac{\sum_{j=1}^{k} S_j(\frac{I_j}{S_j} - (\frac{I}{S})_j^{SS} - \sum_{j=1}^{n} w_j(\frac{I_j}{S_j} - (\frac{I}{S})_j^{SS})) - \sum_{j=n-k+1}^{n} S_j(\frac{I_j}{S_j} - (\frac{I}{S})_j^{SS} - \sum_{j=1}^{n} w_j(\frac{I_j}{S_j} - (\frac{I}{S})_j^{SS}))}{Sales}$$

where S_j is the sales of plant j, w_j is the sales of plant j divided by the firm's total sales, I_j is the capital expenditures of plant j, $(I/S)_j^{SS}$ is the average investment-to-sales ratio of all plants in the same three-digit SIC industry, and Sales is the total sales of the firm. For $j = 1...k$, the firm's plants operate in an industry with median Q greater than the firm's sales-weighted average Q, and $j = (n-k+1)...n$ indicates that the firm's plants have an industry median Q less than the firm's sales-weighted average Q. ΔRINV measures a target firm's change in investment efficiency.

I now estimate Eq. (10) by using ΔWage, ΔEmployment, ΔCapex, and ΔRINV instead of ΔTFP and report estimation results in Table 11. Results in the first two columns indicate that cutbacks on wage and capital expenditure are both positively related to the combined CAR, albeit statistically significant only at the 10% level. The biggest driver of combined CAR comes from ΔRINV. The coefficient of 0.202 in Column 2 implies that acquiring a target with ΔRINV at the 25th percentile yields 0.061 percentage points higher combined CAR than acquiring a target with ΔRINV at the

75th percentile. This estimate suggests that ΔRINV accounts for about two-thirds of the total effects of ΔTFP on the combined CAR. In comparison, an inter-quartile change in ΔCapex increases the combined CAR by only 0.017 percentage points.

In Columns 3 and 4, I report regression results using targets' premiums as dependent variables. The pattern of results resembles the pattern in Columns 1 and 2. Higher premiums are offered to targets with greater cutbacks on wages and capital expenditures, and targets with greater improvements in investment efficiency receive higher premiums. Again, the coefficient on ΔRINV is much greater than other explanatory variables. I find no significant relation between acquirer CAR and any of the components, as evidenced from the last two columns in Table 11.

In sum, my results indicate that the effects of ΔTFP on merger announcement returns are mainly driven by investment efficiency gains, not sheer cutbacks on wages and capital expenditures.

5.5 *Robustness checks and additional analysis*

I perform several robustness checks. First, in the above results, I consider only the TFP changes that occur between T−1 and T+1 when I calculate ΔTFP. This two-year window might be too short because some of the changes in productivity could show up later, as shown in the previous sections. I construct another variable $\Delta TFP_{(-1,+2)}$ to measure TFP changes from T−1 to T+2. In Table 12, I report results when I use $\Delta TFP_{(-1,+2)}$ instead of $\Delta TFP_{(-1,+1)}$. The results are qualitatively similar to those in Tables 8, 9, and 10. $\Delta TFP_{(-1,+2)}$ has a significantly positive effect on combined CAR and offer premium, but not on acquirer CAR. $\Delta TFP_{(-1,+2)}$ has a statistically significant positive effect on combined CAR and acquirer CAR, especially when the target is relatively larger compared with acquirer. The point estimates of $\Delta TFP_{(-1,+2)}$ are generally smaller than those of $\Delta TFP_{(-1,+1)}$. For instance, an inter-quartile increase in target's $\Delta TFP_{(-1,+2)}$ is associated with 6 percentage points increase in offer premiums (relative to 8 percentage points when I use $\Delta TFP_{(-1,+1)}$). An inter-quartile increase in target's $\Delta TFP_{(-1,+2)}$ is associated with 1.6 percentage points increase in combined CAR when target is half the size of acquirer (relative to 2.5 percentage points when I use $\Delta TFP_{(-1,+1)}$). These results are probably because of a higher risk when the market evaluates projects in a longer time span.

Second, as shown in Subsection 4.5, in a relatively smaller sample, I find that acquirers' incumbent plants also experience net improvements in TFP. It is, therefore, interesting to see whether announcement returns embed information regarding productivity changes for acquirers' incumbent plants. I

construct a similar variable *Acquirer's*ΔTFP to measure the FE TFP changes for acquirer's incumbent plants between T−1 and T+1 and examine whether acquirer's ΔTFP affects announcement returns. The results are reported in Table 13. First notice that the effects of target's ΔTFP on announcement returns remain robust to the inclusion of acquirer's ΔTFP. After controlling for target's ΔTFP, acquirer's ΔTFP does not seem to affect the announcement returns, either for the combined firms or the acquiring firms themselves. While the signs on *Acquirer's*ΔTFP are all positive, the large standard errors make the point estimates statistically insignificant. This evidence indicates that the announcement returns embed little information about the productivity changes from acquirers' incumbent plants.

In the above analysis, I calculate the target firm's ΔTFP based on all plants that are retained by acquirers after a takeover. As shown in Subsection 4.2, a portion of the target plants are sold to other firms. As a placebo test, I also run the regressions using a pseudo ΔTFP based on plants that are sold by acquirers to other firms. In untabulated results, I find that changes in TFP for those sold plants are not associated with combined CAR, offer premium, or acquirer CAR.

While the above results demonstrate that announcement returns reflect underlying efficiency improvements, some argue that wealth gains to shareholders can simply represent wealth redistribution from other stakeholders. Shleifer and Summers (1988) argue that takeovers are in part motivated by the opportunity to renege on implicit labor contracts. Empirically, in a sample of 413 takeovers between 1981 and 1988, Pontiff, Shleifer, and Weisbach (1990) report that pension funds are more likely to be reversed following a hostile takeover, but these pension reversions amount to only a small portion of takeover premiums.

I examine the wealth redistribution hypothesis from the perspective of target employees' fringe benefits. In the manufacturing census, employees' fringe benefits include pension plan costs, health insurance, and unemployment benefits. I construct a variable ΔFringe Benefits to measure percentage change in target's cost of fringe benefits from pre- to post-takeover and regress announcement returns on ΔFringe Benefits. If the gains to shareholders are wealth redistribution from employees' fringe benefits, the coefficient on ΔFringe Benefits should be negative. Regression results are presented in Table 14. I include ΔFringe Benefits$_{(-1,+1)}$ in odd columns and ΔFringe Benefits$_{(-1,+2)}$ in even columns (to account for the fact that cuts in fringe benefits could occur later). The results show that post-takeover changes in target employees' fringe benefits are not associated with combined CAR, offer premium, or acquirer CAR, in a statistically significant fashion. These results suggest that the gains

to shareholders are not just wealth redistributions from target employees' fringe benefits.

6 Conclusions

In this paper, I analyze how takeovers create value. I address this question in two steps. First, by examining changes to outputs and inputs after a takeover, I identify the detailed channels through which the acquiring firm improves the target firm's total factor productivity. Second, by establishing a direct relation between a target firm's post-takeover productivity changes and takeover announcement returns, I provide some micro-level evidence to understand the stock market revaluation of the merging firms.

I analyze a sample of 1,430 mergers using plant-level data from the US Bureau of Census. To evaluate the real effects of takeovers, I construct a group of control plants that are comparable to target plants in terms of industry, size, and productivity. Using a difference-in-differences approach, I find that while the target plants undergo significant cutbacks in capital expenditures, wage rates, and employment, the level of output remains virtually unchanged. Employee layoffs are concentrated mostly among nonproduction workers. Also, the sensitivities of targets' capital expenditures to measures of investment opportunities rise significantly after the takeover, especially when the acquirer was a conglomerate firm and ran an active internal capital market before the acquisition.

The changes in targets' productivity help explain the cross-sectional differences in merging firms' announcement returns. I find that improvements in productivity are associated with higher combined announcement returns. The effect of improvements in productivity on combined announcement return is statistically robust and economically significant, especially when the target is relatively larger compared with the acquirer. Moreover, targets with greater future improvements in productivity receive higher offer premiums from acquirers. I also find that target's improvements in productivity have a positive effect on acquirers' announcement returns only when the target is relatively large compared with the acquirer. My analysis uncovers a significant relation between the target's productivity improvements and the announcement returns.

Overall, this study shows that takeovers are an efficient way to reallocate production assets to management that can better use them. It highlights the interconnections between productivity and stock returns in the context of takeovers.

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Table 1: **Number of deals and basic deal characteristics**

The table describes the basic characteristics of the takeover sample in this study. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM/CMF, maintained at the U.S. Census Bureau. Reported are the number of mergers completed each year, the mean and standard deviation of deal value (in millions, reported in SDC)and the mean and standard deviation of the three-day cumulative abnormal return (CAR) around deal announcements, for the target and the acquirer. CAR is estimated using a market model adjusted for CRSP value-weighted market index using the (-205, -6) event window.

		Deal value		Target $CAR_{(-1,+1)}$		Acquirer $CAR_{(-1,+1)}$	
Year	N	Mean	S.D.	Mean	S.D.	Mean	S.D.
1981	15	322.8	457.9	0.251	0.332	−0.004	0.052
1982	19	219.4	285.8	0.162	0.148	−0.013	0.047
1983	34	157.1	239.3	0.110	0.149	0.016	0.122
1984	48	285.3	415.3	0.138	0.171	−0.009	0.059
1985	60	747.6	1,297.6	0.186	0.176	−0.014	0.052
1986	103	524.3	1,163.8	0.167	0.177	0.004	0.051
1987	76	468.6	733.9	0.216	0.187	0.019	0.058
1988	124	504.3	1392.5	0.228	0.235	−0.009	0.087
1989	85	969.3	3662.5	0.213	0.209	−0.007	0.070
1990	53	506.4	917.9	0.272	0.269	−0.020	0.047
1991	24	1,178.1	2,385.8	0.345	0.385	0.018	0.087
1992	16	195.4	262.1	0.292	0.202	0.035	0.081
1993	40	312.0	666.6	0.287	0.213	−0.007	0.056
1994	42	984.9	2,209.5	0.208	0.160	−0.005	0.066
1995	60	790.3	1,646.6	0.224	0.177	−0.012	0.058
1996	83	1,053.8	2,754.4	0.226	0.237	−0.001	0.069
1997	85	746.4	1,625.6	0.236	0.202	0.016	0.068
1998	126	2,226.3	7,936.6	0.265	0.230	0.010	0.078
1999	128	1,444.3	3,009.8	0.262	0.212	0.003	0.086
2000	112	3,095.4	10,260.4	0.302	0.286	−0.015	0.087
2001	62	2,239.3	5,782.7	0.345	0.534	−0.031	0.097
2002	35	1,919.5	5049.8	0.152	0.225	−0.023	0.094
Total	1430	949.6	4,480.34	0.233	0.248	−0.002	0.074

Table 2: **Descriptive statistics for target firm and plant output and employment**

The table reports summary statistics for some variables at the target firm and plant level. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. T is the year when the merger is completed. Fixed Effects Total factor productivity (FE TFP) is the residual from estimating a log linear Cobb-Douglas production function at the plant level, in which one regresses the value of output on total payroll, capital stock, and material cost controlling for plant fixed effects. Translog TFP is the residual from estimating a translog type of production function, in which one regresses the value of output on total payroll, capital stock, and material cost, as well as their square terms and cross products (all in logs) controlling for plant fixed effects. Employment is the total number of employees. Output is the total value of shipment adjusted for industry deflators.

Panel A: Firm level

		Plants per firm		Total employment		Total output	
Timing	N	Mean	SD	Mean	SD	Mean	SD
T − 3	1,272	7.38	14.46	2,729.4	6,629.7	631.5	3,223.8
T − 2	1,305	7.54	15.25	2,700.5	6,530.7	645.1	3,277.5
T − 1	1,430	8.09	16.21	2,531.7	6,276.8	635.1	3,367.6
T	1,275	7.51	15.25	2,588.6	6,365.9	664.6	3,065.0
T + 1	1,228	7.44	14.87	2,540.7	6,221.8	673.9	2,928.2
T + 2	1,169	7.22	14.42	2,539.2	6,347.1	711.2	3,024.8
T + 3	1,139	6.92	13.60	2,427.7	5,797.9	739.2	3,139.6

Panel B: Plant level

		FE TFP		Translog TFP		Output	
Timing	N	Mean	SD	Mean	SD	Mean	SD
T − 3	9,045	0.041	0.336	0.086	0.397	75.9	295.7
T − 2	9,465	0.031	0.324	0.077	0.402	75.7	299.7
T − 1	11,129	0.039	0.348	0.080	0.437	70.9	301.1
T	9,201	0.053	0.340	0.087	0.466	78.2	261.1
T + 1	8,742	0.061	0.339	0.094	0.485	80.6	258.3
T + 2	8,129	0.075	0.353	0.107	0.497	86.6	279.8
T + 3	7,591	0.095	0.355	0.128	0.479	96.2	331.7

Table 3: **Effects of takeovers on outputs and inputs**

The table reports estimates of a plant-level difference-in-differences regression of $\text{Log}(Y_{it}) = \alpha + \beta_1 \text{After}_t + \beta_2(\text{After}_t * \text{Target}_i) + \text{Plant \& Year FE} + \epsilon_{it}$. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Output is the total value of shipment. Investment is the total capital expenditure. Material is the sum of cost of materials, parts, energy, and electricity costs. Wage is calculated as total salaries divided by total number of employees. Employment is the total number of employees. All these variables are in logs. The data are an unbalanced six-year panel (T−3, T−2, T−1, T+1, T+2, and T+3) for target and control plants. T, the year when the merger is completed, is dropped from the analysis. After is a dummy variable that equals one for year T+1, T+2, and T+3 and zero for T−1, T−2, and T−3. Target is a dummy variable that equals one if a plant's parent firm is acquired at year T and zero if a plant is a control plant. The control sample is constructed by selecting plants that are comparable to target plants in terms of industry, size, and total factor productivity prior to the takeover. Heteroskedastic-robust standard errors clustered at firm level are in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	Output	Investment	Material	Wage	Employment
	(1)	(2)	(3)	(4)	(5)
After	−0.022***	−0.093***	−0.016***	0.003	−0.014***
	(0.004)	(0.014)	(0.005)	(0.002)	(0.004)
After * Target	−0.002	−0.124***	−0.009**	−0.005*	−0.021***
	(0.006)	(0.034)	(0.004)	(0.003)	(0.006)
Plant Fixed Effect	Yes	Yes	Yes	Yes	Yes
Year Fixed Effect	Yes	Yes	Yes	Yes	Yes
Number of observations	171,754	171,754	171,754	171,754	171,754
Adj. R^2	0.89	0.70	0.89	0.77	0.92

	Production worker		Nonproduction worker	
	Wage	Employment	Wage	Employment
	(1)	(2)	(3)	(4)
After	0.007***	−0.019***	−0.029*	−0.002
	(0.002)	(0.004)	(0.015)	(0.006)
After * Target	−0.002	−0.014**	−0.008	−0.059***
	(0.004)	(0.007)	(0.013)	(0.013)
Plant Fixed Effects	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes
Number of observations	171,754	171,754	171,754	171,754
Adj. R^2	0.75	0.91	0.43	0.85

Table 4: **Sensitivity of investments to Q before and after takeovers**

The table reports estimates of Eqs. (8) and (9) as given in the text. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly-listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. The dependent variable Investment is capital expenditures divided by sales. Industry Q is the median Q in the industry in which the plant operates. Industry investments is the average industry-wide investment calculated from census database. Profit ratio is calculated as the value of shipments net of total payroll, material costs and capital expenditures, divided by sales. The data are an unbalanced six-year panel (T−3, T−2, T−1, T+1, T+2, and T+3) for target and control plants separately. T, the year when the merger is completed, is dropped out from the analysis. After is a dummy variable that equals one for year T+1, T+2, and T+3 and zero for T−1, T−2, and T−3. The control sample is constructed by selecting plants that are comparable to target plants in terms of industry, size, and total factor productivity prior to the takeover. Heteroskedastic-robust standard errors clustered at firm level are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level.

	Dependent variable: Investment/Sale					
	Target Sample			Control Sample		
	(1)	(2)	(3)	(4)	(5)	(6)
After	−0.022***	−0.019***	−0.018***	−0.002	0.012	0.009
	(0.005)	(0.004)	(0.005)	(0.003)	(0.009)	(0.005)
Industry Q	0.027**		0.021*	0.024**		0.025**
	(0.013)		(0.014)	(0.012)		(0.013)
Industry Q * After	0.018**		0.020***	−0.031		0.011
	(0.009)		(0.008)	(0.022)		(0.019)
Industry Investments		0.012***			0.011	
		(0.003)			(0.013)	
Industry Investments * After		0.007***			0.015*	
		(0.002)			(0.008)	
Profits			0.033*			0.037**
			(0.017)			(0.018)
Profits * After			0.073			0.091
			(0.064)			(0.069)
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	50,133	50,133	50,133	121,621	121,621	121,621
Adj. R^2	0.87	0.86	0.89	0.84	0.85	0.88

Table 5: **Investments cutbacks before and after takeovers: heterogeneous effects**

The table reports estimates of Eq. (7) for investments controlling for additional characteristics of merger and acquirer. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Investment is the total capital expenditure. The data are an unbalanced six-year panel (T−3, T−2, T−1, T+1, T+2, and T+3) for target and control plants. T, the year when the merger is completed, is dropped from the analysis. After is a dummy variable that equals one for year T+1, T+2, and T+3 and zero for T−1, T−2, and T−3. Target is a dummy variable that equals one if a plant's parent firm is acquired at year T and zero if a plant is a control plant. The control sample is constructed by selecting plants that are comparable to target plants in terms of industry, size, and total factor productivity prior to the takeover. Related, Conglomerate, Diversity, Reallocate, and 1/Herfindahl are measures of acquirer's internal capital market intensity defined in Subsection 4.3. Heteroskedastic-robust standard errors clustered at firm level are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Dependent variable: Investment						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
After	−0.081***	−0.080***	−0.078***	−0.076***	−0.083***	−0.085***	−0.079***
	(0.015)	(0.014)	(0.016)	(0.019)	(0.017)	(0.018)	(0.019)
After * Target			−0.094***	−0.097***	−0.088***	−0.131***	−0.122***
			(0.037)	(0.039)	(0.038)	(0.040)	(0.041)
After * Target * Related	−0.136***						
	(0.041)						
After* Target * (1-Related)	−0.121***						
	(0.038)						
After* Target * Conglomerate		−0.134***					
		(0.029)					
After * Target * (1-Conglomerate)		−0.078					
		(0.055)					
After * Target * Diversity			−0.496***				
			(0.128)				
After * Target * Reallocate				−0.472**			
				(0.129)			
After * Target * 1/Herfindahl					−0.545**		
					(0.208)		
After * Target * Acquirer Size						0.725*	
						(0.396)	
After * Target * Acquirer M/B							0.212
							(0.597)
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	171,754	150,237	134,381	132,756	134,484	137,986	135,981
Adj. R^2	0.73	0.71	0.72	0.73	0.71	0.70	0.74

Table 6: **Sensitivity of investments to Q before and after takeovers: heterogeneous effects**

The table reports estimates of Eq. (8) controlling for additional characteristics of merger and acquirer. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Related, Conglomerate, Diversity, Reallocate, and 1/Herfindahl, are measures of acquirer's internal capital market intensity defined in Subsection 4.3. The dependent variable is investments divided by sales. Industry Q is the median Q in the three-digit SIC industry in which the plant operates. The data are an unbalanced six-year panel (T−3, T−2, T−1, T+1, T+2, and T+3) for target plants only. Heteroskedastic-robust standard errors clustered at firm level are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Dependent variable: Investment						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
After	−0.021***	−0.020***	−0.018***	−0.016*	−0.013**	−0.015*	−0.019**
	(0.005)	(0.004)	(0.006)	(0.009)	(0.007)	(0.008)	(0.009)
Q	0.024**	0.019*	0.025*	0.022**	0.021*	0.030***	0.031***
	(0.011)	(0.012)	(0.014)	(0.010)	(0.013)	(0.012)	(0.009)
After * Q			0.014**	0.014*	0.018**	0.021**	0.022**
			(0.007)	(0.009)	(0.008)	(0.010)	(0.011)
After * Q * Related	0.014*						
	(0.008)						
After * Q * (1-Related)	0.020***						
	(0.007)						
After * Q * Conglomerate		0.024**					
		(0.011)					
After * Q * (1-Conglomerate)		0.007					
		(0.005)					
After * Q * Diversity			0.091***				
			(0.028)				
After * Q * Reallocate				0.072**			
				(0.033)			
After * Q * 1/Herfindahl					0.045*		
					(0.028)		
After * Q * Acquirer Size						0.327	
						(0.417)	
After * Q * Acquirer M/B							0.819
							(0.722)
Plant Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	50,133	50,133	39,311	38,905	37,122	37,187	36,760
Adj. R^2	0.87	0.86	0.82	0.83	0.81	0.77	0.79

Table 7: **Summary statistics of key variables in announcement returns analysis**

The table reports mean, standard deviation, and pairwise correlation for several variables. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Offer Premium is the ratio of the transaction value to target's equity value 30 days before announcement minus one. Acquirer CAR is the three-day cumulative abnormal returns (CAR), estimated using a market model, for the acquiring firm. Combined CAR is equity value weighted average of target's CAR and acquirer's CAR. TFP is constructed as an output-weighted sum of all of total factor productivity of the target plants retained by acquirers. $\Delta\text{TFP} = \text{TFP}_{\text{T+1}} - \text{TFP}_{\text{T-1}}$.

		N	Mean	SD	(1)	(2)	(3)	(4)
(1)	ΔTFP	1,226	0.041	0.259	1			
(2)	Offer Premium	1,226	0.782	1.613	0.030	1		
(3)	Acquirer CAR	660	-0.002	0.074	0.005	0.001	1	
(4)	Combined CAR	660	0.032	0.094	0.047	0.001	0.058	1

Table 8: **Determinants of combined cumulative abnormal returns (CAR)**

The table reports estimates of a firm-level OLS regression explaining the combined CAR. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly-listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. CAR is equity value weighted average of target's CAR and acquirer's CAR, which are estimated using a market model adjusting for CRSP value-weighted market index. TFP is constructed as an output-weighted sum of all of FE-TFP of the target plants retained by acquirers. $\Delta\text{TFP} = \text{TFP}_{\text{T+1}} - \text{TFP}_{\text{T-1}}$. RelaSize is the ratio of the target's market value to the sum of target's and acquirer's market value. All Cash is a dummy variable that equals one if the entire transaction value was paid by cash and zero otherwise. Some Stock is a dummy variable that equals one if at least some portion of the transaction was paid by stock and zero otherwise. Related is a dummy variable that equals one if target and acquirer share the same primary two-digit SIC code. Tender Offer is a dummy variable that equals one if the acquisition is a tender offer. Log(Aquirer/Target Assets) is the logarithm of a firm's book value of assets. Market-to-book is the ratio of market value of assets to book value of assets. A constant plus year and industry (2-digit SIC) fixed effects for both target and acquirer are included, but not reported. Heteroskedastic-robust standard errors are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Combined CAR					
	(1)	(2)	(3)	(4)	(5)	(6)
ΔTFP	0.027*	0.033**	0.036**	−0.005	−0.018	−0.019
	(0.015)	(0.015)	(0.015)	(0.018)	(0.018)	(0.019)
RelaSize	0.175***	0.175***	0.187***	0.176***	0.177***	0.168***
	(0.014)	(0.014)	(0.018)	(0.014)	(0.010)	(0.023)
RelaSize * ΔTFP				0.218***	0.299***	0.298***
				(0.052)	(0.065)	(0.065)
Related	0.017**	0.015*	0.025***	0.018**	0.015*	0.024***
	(0.008)	(0.008)	(0.008)	(0.007)	(0.008)	(0.008)
All Cash	−0.003	−0.009	−0.005	−0.004	−0.011	−0.006
	(0.009)	(0.010)	(0.011)	(0.009)	(0.010)	(0.011)
Some Stock	−0.027***	−0.023***	−0.019**	−0.027***	−0.023***	−0.020**
	(0.009)	(0.011)	(0.012)	(0.009)	(0.011)	(0.011)
Tender Offer	0.016*	0.018*	0.011	0.015*	0.017*	0.010
	(0.008)	(0.009)	(0.010)	(0.008)	(0.009)	(0.010)
Log(Acquirer Asset)		−0.007**	−0.007**		−0.004*	−0.006**
		(0.003)	(0.003)		(0.002)	(0.003)
Acquirer Market-to-book		−0.001	−0.001		−0.001	−0.001
		(0.004)	(0.004)		(0.003)	(0.004)
Log(Target Asset)			−0.008**			−0.006
			(0.004)			(0.004)
Target Market-to-Book			−0.007*			−0.007**
			(0.004)			(0.003)
Number of observations	660	459	395	660	459	395
Adj. R^2	0.24	0.25	0.25	0.29	0.30	0.31

Table 9: **Determinants of offer premium**

The table reports estimates of firm-level OLS regression explaining the target's offer premium. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Offer Premium is the ratio of transaction value to target's market value 30 trading days before announcement minus one. TFP is constructed as an output-weighted sum of all of total factor productivity of the target plants retained by acquirers. ΔTFP = $\text{TFP}_{\text{T+1}} - \text{TFP}_{\text{T-1}}$. All Cash is a dummy variable that equals one if the entire transaction value was paid by cash and zero otherwise. Some Stock is a dummy variable that equals one if at least some portion of the transaction was paid by stock and zero otherwise. Related is a dummy variable that equals one if target and acquirer share the same primary two-digit SIC industry. Tender Offer is a dummy variable that equals one if the acquisition is a tender offer. 52-W High Price is the ratio of target's highest price in the previous 52 weeks to target's price 30 days before announcement. Log(Assets) is the logarithm of a firm's book value of assets. Market-to-book is the ratio of market value to the book value of assets. A constant plus year and industry (two-digit SIC) fixed effects for both target and acquirer are included, but not reported. Heteroskedastic-robust standard errors are in parentheses. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Offer Premium					
	(1)	(2)	(3)	(4)	(5)	(6)
ΔTFP	0.319**	0.330**	0.297**	0.455*	0.330**	0.306**
	(0.122)	(0.162)	(0.138)	(0.249)	(0.162)	(0.139)
RelaSize		−0.928***	−1.184***	−0.910***	−0.923***	−1.196***
		(0.236)	(0.324)	(0.281)	(0.234)	(0.322)
RelaSize * ΔTFP				0.069	0.071	0.087
				(0.636)	(0.612)	(0.618)
Related	0.287**	0.325*	0.359**	0.322*	0.320**	0.358**
	(0.142)	(0.186)	(0.162)	(0.187)	(0.146)	(0.162)
All Cash	−0.574***	−0.573***	−0.594***	−0.572***	−0.572***	−0.594***
	(0.179)	(0.204)	(0.229)	(0.178)	(0.204)	(0.229)
Some Stock	−0.187	−0.181	−0.520**	−0.486	−0.434**	−0.520**
	(0.175)	(0.225)	(0.233)	(0.253)	(0.205)	(0.234)
Tender Offer	0.410**	0.534***	0.434**	0.538**	0.435**	0.434**
	(0.123)	(0.213)	(0.183)	(0.213)	(0.162)	(0.183)
52-W High price	0.531***	0.703***	0.785***	0.701***	0.751***	0.786***
	(0.156)	(0.241)	(0.263)	(0.240)	(0.248)	(0.263)
Log(Target Asset)	0.057*	0.161**	0.077*		0.160**	0.079*
	(0.033)	(0.078)	(0.045)		(0.078)	(0.045)
Target Market-to-Book	−0.129***	−0.141**	−0.170***		−0.141**	−0.170***
	(0.044)	(0.058)	(0.059)		(0.058)	(0.059)
Log(Acquirer Asset)			−0.006			−0.008
			(0.052)			(0.053)
Acquirer Market-to-Book			−0.049			−0.048
			(0.056)			(0.056)
Number of observations	1,004	560	395	660	560	395
Adj. R^2	0.10	0.21	0.24	0.10	0.21	0.24

Table 10: **Determinants of acquirer cumulative abnormal returns (CAR)**

The table reports estimates of a firm-level OLS regression explaining the acquirer's CAR. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. CAR is the three-day cumulative abnormal return for the acquirer using a market model adjusted for CRSP value-weighted market index. TFP is constructed as an output-weighted sum of all of total factor productivity of the target plants retained by acquirers. $\Delta\text{TFP} = \text{TFP}_{\text{T+1}} - \text{TFP}_{\text{T-1}}$. All Cash is a dummy variable that equals one if the entire transaction value was paid by cash and zero otherwise. Some Stock is a dummy variable that equals one if at least some portion of the transaction was paid by stock and zero otherwise. Tender Offer is a dummy variable that equals one if the acquisition is a tender offer. Log(Assets) is the logarithm of a firm's book value of assets. Market-to-book is the ratio of market value to the book value of assets. Heteroskedastic-robust standard errors are in parentheses. A constant plus year and industry (two-digit SIC) fixed effects are included, but not reported. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Acquirer CAR					
	(1)	(2)	(3)	(4)	(5)	(6)
ΔTFP	0.010	0.021*	0.019	0.014	0.005	0.005
	(0.012)	(0.012)	(0.013)	(0.016)	(0.015)	(0.017)
RelaSize	−0.027*	0.028	0.026	−0.031*	0.021	0.027
	(0.014)	(0.019)	(0.015)	(0.017)	(0.015)	(0.016)
RelaSize * ΔTFP				0.097*	0.116**	0.108**
				(0.054)	(0.057)	(0.054)
Related	0.001	0.006	0.010	0.001	0.006	0.010
	(0.006)	(0.007)	(0.007)	(0.006)	(0.007)	(0.007)
All Cash	−0.001	−0.001	−0.007	−0.001	−0.002	−0.008
	(0.008)	(0.008)	(0.009)	(0.008)	(0.008)	(0.009)
Some Stock	−0.020***	−0.017***	−0.017*	−0.020***	−0.017*	−0.017**
	(0.008)	(0.009)	(0.009)	(0.008)	(0.009)	(0.010)
Tender Offer	0.004	0.011	0.011	0.005	0.010	0.011
	(0.007)	(0.008)	(0.009)	(0.007)	(0.008)	(0.009)
Log(Acquirer Asset)		−0.006***	−0.009***		−0.006***	−0.009***
		(0.002)	(0.003)		(0.002)	(0.003)
Acquirer Market-to-Book		−0.001	−0.001		−0.001	−0.001
		(0.003)	(0.003)		(0.003)	(0.003)
Log(Target Asset)			−0.004			−0.004
			(0.003)			(0.003)
Target Market-to-Book			−0.004			−0.004
			(0.003)			(0.003)
Number of observations	660	459	395	660	459	395
Adj. R^2	0.02	0.06	0.08	0.02	0.06	0.08

Table 11: **Breaking down the effect of ΔTFP**

The table reports estimates of a firm-level OLS regression explaining the announcement returns. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Cumulative abnormal return (CAR) is the three-day cumulative abnormal return for the acquirer using a market model adjusted for CRSP value-weighted market index. X is constructed as an output-weighted sum of all X of the target plants retained by acquirers. $\Delta X = X_{T+1} - X_{T-1}$, where X includes average wages, total employment, capital expenditures, and relative investment ratio (RINV). RINV is a measure of overall investment efficiency defined as the sales-weighted sum of investments in high-Q plants minus investments in low-Q plants. A higher RINV corresponds to a greater investment efficiency. Deal characteristics include All Cash, Some Stock, Related, and Tender Offer. Acquirer(target) characteristics include acquirer's(target)'s Log(Assets) and Market-to-book ratio. Heteroskedastic-robust standard errors are in parentheses. A constant plus year and industry (two-digit SIC) fixed effects for both target and acquirer are included, but not reported. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Combined CAR		Premium		Acquirer CAR	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔWage	−0.028*	−0.025*	−0.309***	−0.352**	−0.012	−0.007
	(0.016)	(0.015)	(0.117)	(0.171)	(0.012)	(0.015)
ΔEmployment	−0.017	0.018*	0.034	0.041	−0.007	−0.023
	(0.015)	(0.011)	(0.011)	(0.016)	(0.029)	(0.025)
ΔCapex	−0.029*	−0.027*	−0.263**	−0.239**	0.038	0.039
	(0.019)	(0.014)	(0.129)	(0.122)	(0.035)	(0.044)
ΔRINV	0.214**	0.202**	0.911***	0.872***	0.103	0.092
	(0.101)	(0.098)	(0.216)	(0.341)	(0.081)	(0.078)
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer characteristics		Yes		Yes		Yes
Target characteristics		Yes		Yes		Yes
Number of observations	660	395	1004	395	660	395
Adj. R^2	0.04	0.24	0.04	0.24	0.04	0.24

Table 12: **Determinants of announcement returns using ΔTFP$_{(-1,+2)}$**

The table reports estimates of a firm-level OLS regression explaining the announcement returns. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. CAR is the three-day cumulative abnormal return for the acquirer using a market model adjusted for CRSP value-weighted market index. TFP is constructed as an output-weighted sum of all of total factor productivity of the target plants retained by acquirers. $\Delta\text{TFP}_{(-1,+2)} = \text{TFP}_{\text{T+2}} - \text{TFP}_{\text{T-1}}$. Deal Characteristics include All Cash, Some Stock, Related, and Tender Offer. Acquirer(target) characteristics include acquirer(target)'s Log(Assets) and Market-to-book ratio. Heteroskedastic-robust standard errors are in parentheses. A constant plus year and industry (two-digit SIC) fixed effects for both target and acquirer are included, but not reported. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Combined CAR		Premium		Acquirer CAR	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta\text{TFP}_{(-1,+2)}$	0.027*	0.013	0.259**	0.271*	0.008	0.001
	(0.014)	(0.015)	(0.123)	(0.139)	(0.010)	(0.013)
RelaSize		0.194***		−0.712***		−0.021
		(0.018)		(0.288)		(0.014)
RelaSize*$\Delta\text{TFP}_{(-1,+2)}$		0.214***		0.079		0.075**
		(0.067)		(0.705)		(0.034)
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer characteristics	Yes	Yes			Yes	Yes
Target characteristics			Yes	Yes		
Number of observations	440	440	532	532	440	440
Adj. R^2	0.04	0.24	0.04	0.24	0.04	0.24

Table 13: **Determinants of announcement returns using acquirer's ΔTFP**

The table reports estimates of a firm-level OLS regression explaining the acquirer's CAR. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. CAR is the three-day cumulative abnormal return for the acquirer using a market model adjusted for CRSP value-weighted market index. TFP is constructed as an output-weighted sum of all of total factor productivity of the target plants retained by acquirers. $\Delta\text{TFP} = \text{TFP}_{\text{T+1}} - \text{TFP}_{\text{T-1}}$. Acquirer$\Delta$TFP is calculated based on acquirer's incumbent plants before and after a takeover. Deal characteristics include All Cash, Some Stock, Related, and Tender Offer. Acquirer(target) characteristics include acquirer(target)'s Log(Assets) and Market-to-book ratio. Heteroskedastic-robust standard errors are in parentheses. Heteroskedastic-robust standard errors are in parentheses. A constant plus year and industry (two-digit SIC) fixed effects for both target and acquirer are included, but not reported. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Combined CAR		Premium		Acquirer CAR	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔTFP	0.036**	0.012	0.387*	0.283**	0.023	−0.004
	(0.018)	(0.023)	(0.205)	(0.015)	(0.015)	(0.020)
RelaSize		0.150***		0.749***		−0.046**
		(0.025)		(0.279)		(0.022)
RelaSize * ΔTFP		0.253**		0.081		0.114**
		(0.113)		(0.712)		(0.048)
Acquirer'sΔTFP	0.016	0.016	0.270	0.019	0.011	0.008
	(0.019)	(0.018)	(0.217)	(0.235)	(0.015)	(0.015)
Relasize * Acquirer'sΔTFP		0.002		−0.003		0.003
		(0.009)		(0.139)		(0.016)
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer characteristics	Yes	Yes			Yes	Yes
Target characteristics			Yes	Yes		
Number of observations	290	290	290	290	290	290
Adj. R^2	0.04	0.24	0.04	0.24	0.04	0.24

Table 14: **Redistribution to shareholders from target employees' fringe benefits**

The table reports estimates of a firm-level OLS regression explaining merger announcement returns. The sample consists of mergers from the SDC M&As database in which the announcement date is between 1981 and 2002, the completion date is within 360 days of the announcement, and the target is a publicly listed firm that can be matched to the ASM and CMF, maintained at the US Census Bureau. Cumulative abnormal returns (CAR) is the three-day cumulative abnormal return estimated using a market model adjusted for CRSP value-weighted market index. ΔFringe Benefits$_{(-1,+1)}$ = Log(Fringe Benefits$_{\text{T+1}}$) − Log(Fringe Benefits$_{\text{T-1}}$), where Fringe Benefits includes target firm's pension costs, health insurance, and unemployment benefits, expressed in 2005 dollars. Deal characteristics include All Cash, Some Stock, Related, and Tender Offer. Acquirer(target) characteristics include acquirer's(target)'s Log(Assets) and Market-to-book ratio. Heteroskedastic-robust standard errors are in parentheses. Heteroskedastic-robust standard errors are in parentheses. A constant plus year and industry (2-digit SIC) fixed effects for both target and acquirer are included, but not reported. ***, **, and * indicate statistical significance at 1%, 5%, and 10% level, respectively.

	Combined CAR		Premium		Acquirer CAR	
	(1)	(2)	(3)	(4)	(5)	(6)
ΔFringe Benefits$_{(-1,+1)}$	−0.003		0.003		0.126	
	(0.008)		(0.006)		(0.102)	
ΔTFP$_{(-1,+1)}$	0.042**		0.301**		0.021*	
	(0.016)		(0.122)		(0.012)	
ΔFringe Benefits$_{(-1,+2)}$		−0.001		−0.006		−0.001
		(0.001)		(0.016)		(0.001)
ΔTFP$_{(-1,+2)}$		0.031**		0.297**		0.006
		(0.014)		(0.133)		(0.010)
Deal characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Acquirer characteristics	Yes	Yes			Yes	Yes
Target characteristics			Yes	Yes		
Number of observations	460	440	1001	951	460	440
Adj. R^2	0.10	0.07	0.09	0.09	0.06	0.06